Genuine Multipartite Entanglement Analysis

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Model
-----
Hubbard ring (1-D, PBC)

System parameters
-----------------
Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 1.0
Interaction (U)   : 1.0
Mode              : uniform
n-body order      : 4
```

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Ground-state properties
-----------------------
E0                : -3.34084762
E1                : -3.29295138

<S2>              : 0.000000
S                 : 0.0000
<Sz>              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
<D> total         : 0.577987
<D> / site        : 0.144497
```

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GME summary
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Extracted n-spin RDM : n = 4

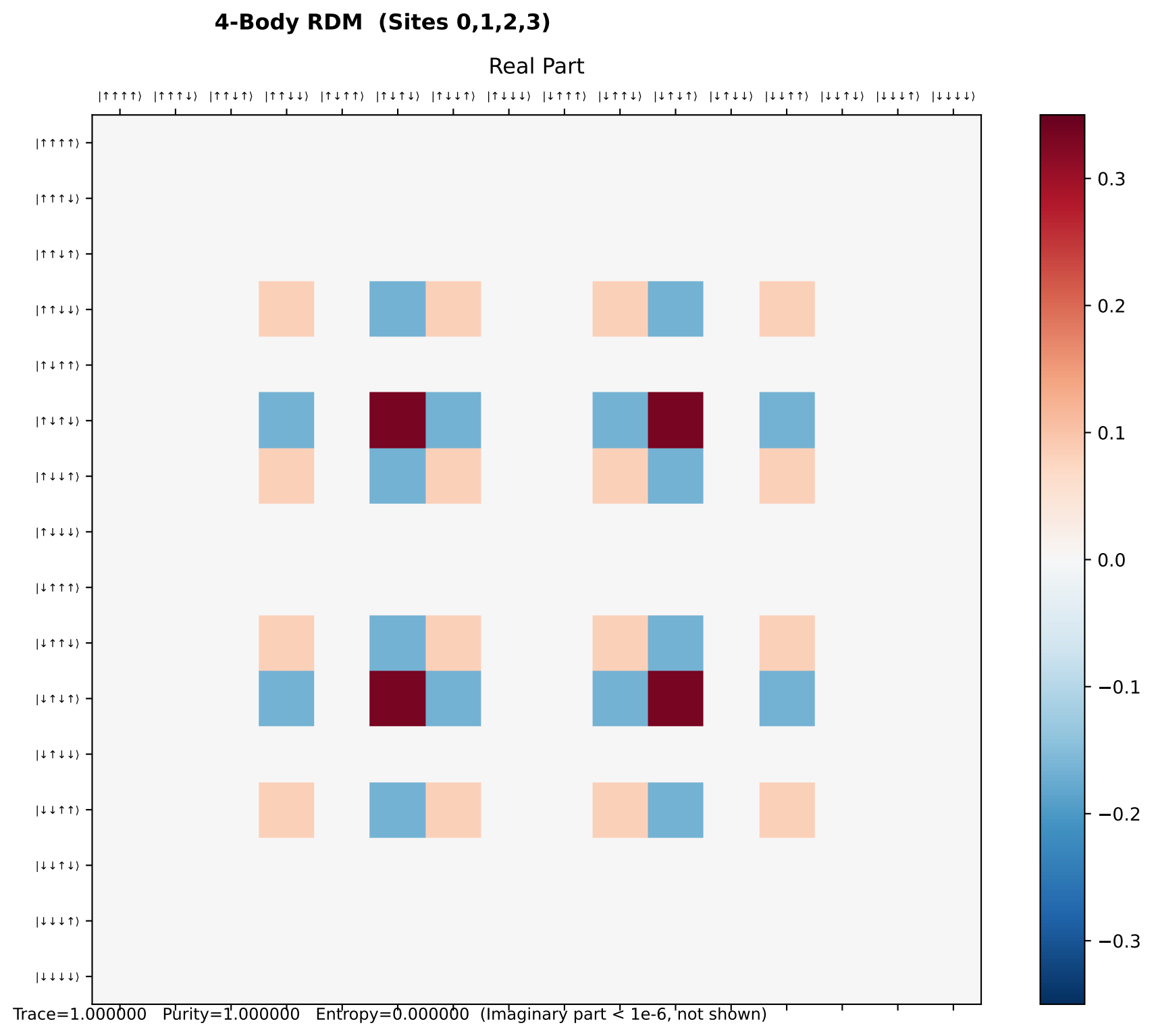
Purity:
  gamma = Tr(rho^2)

Separable cut criterion:
  gamma_A = 1  <->  rho = rho_A x rho_B

C_GME (Troiani et al.):
  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Decision logic:
  gamma=1, all Tr(rho_A^2)<1  -> GME
  gamma=1, some Tr(rho_A^2)=1 -> No GME
  gamma < 1                   -> Mixed

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4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-7.622349e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

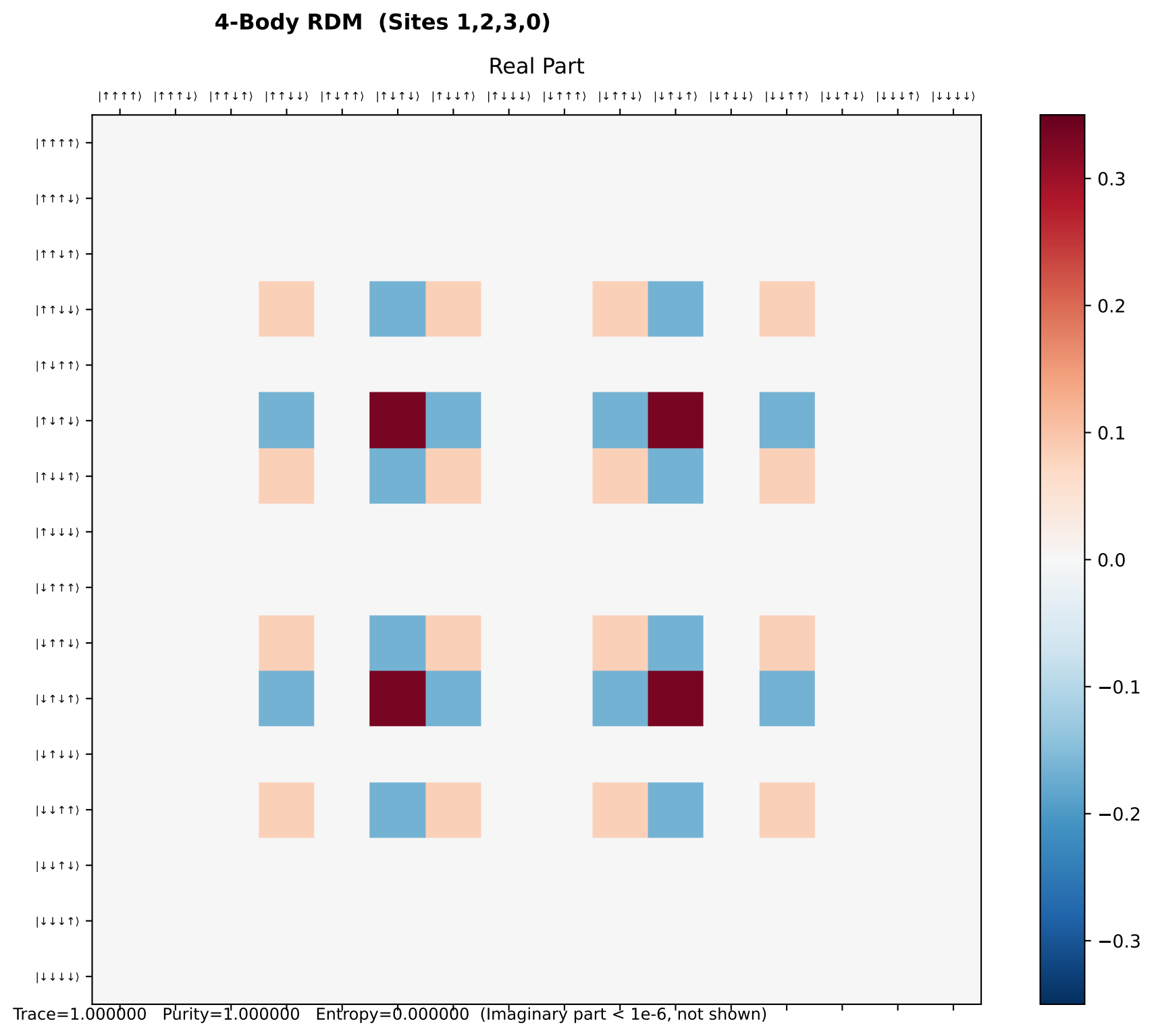
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3)	0.50000000
(0, 1)	(2, 3)	0.58333333
(0, 2)	(1, 3)	0.33333333
(0, 3)	(1, 2)	0.58333333
(0, 1, 2)	(3,)	0.50000000
(0, 1, 3)	(2,)	0.50000000
(0, 2, 3)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.


4-Body RDM (Sites 1,2,3,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-4.334192e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

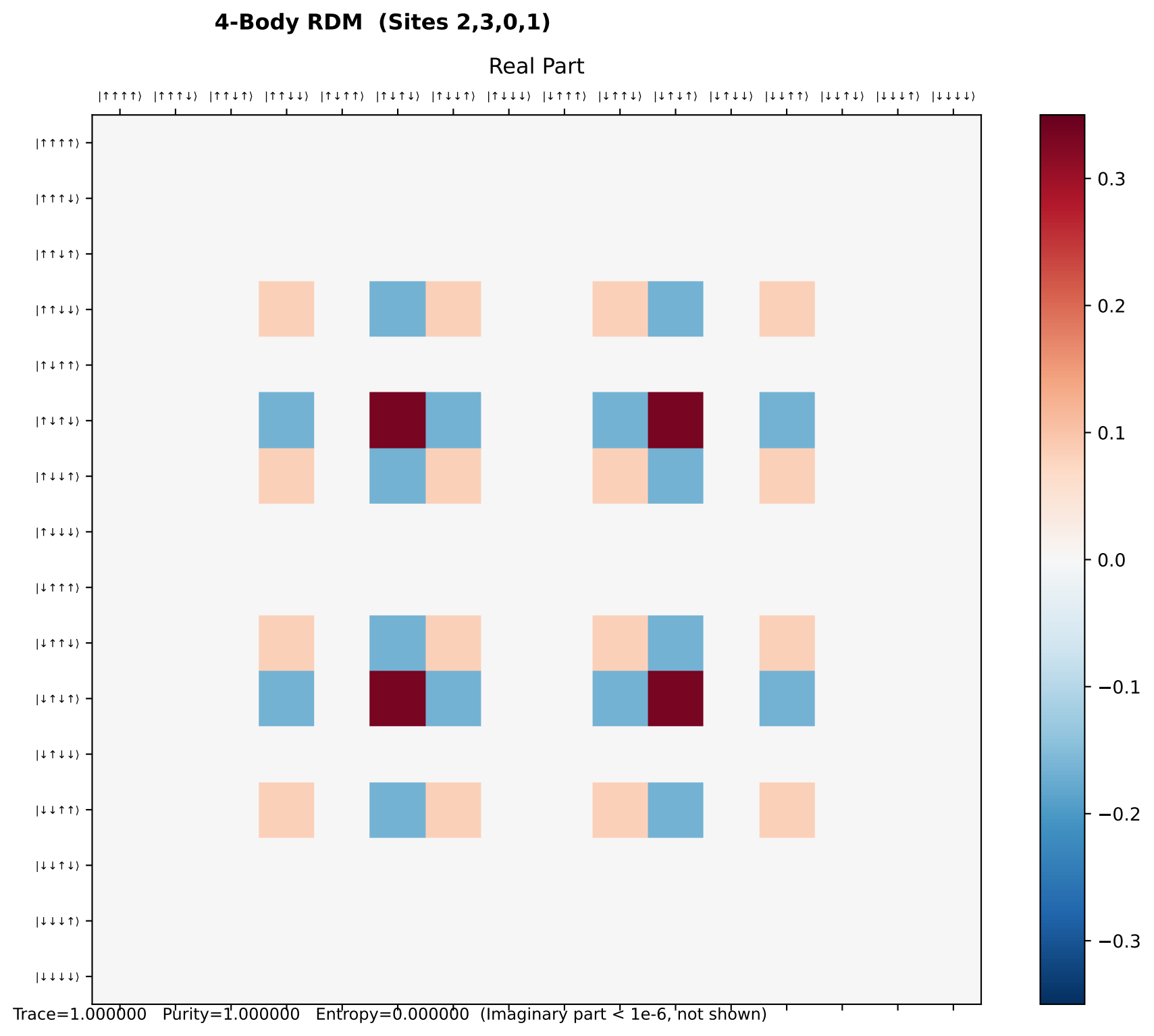
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 0)	0.50000000
(1, 2)	(3, 0)	0.58333333
(1, 3)	(2, 0)	0.33333333
(1, 0)	(2, 3)	0.58333333
(1, 2, 3)	(0,)	0.50000000
(1, 2, 0)	(3,)	0.50000000
(1, 3, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.


4-Body RDM (Sites 2,3,0,1)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-7.228306e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

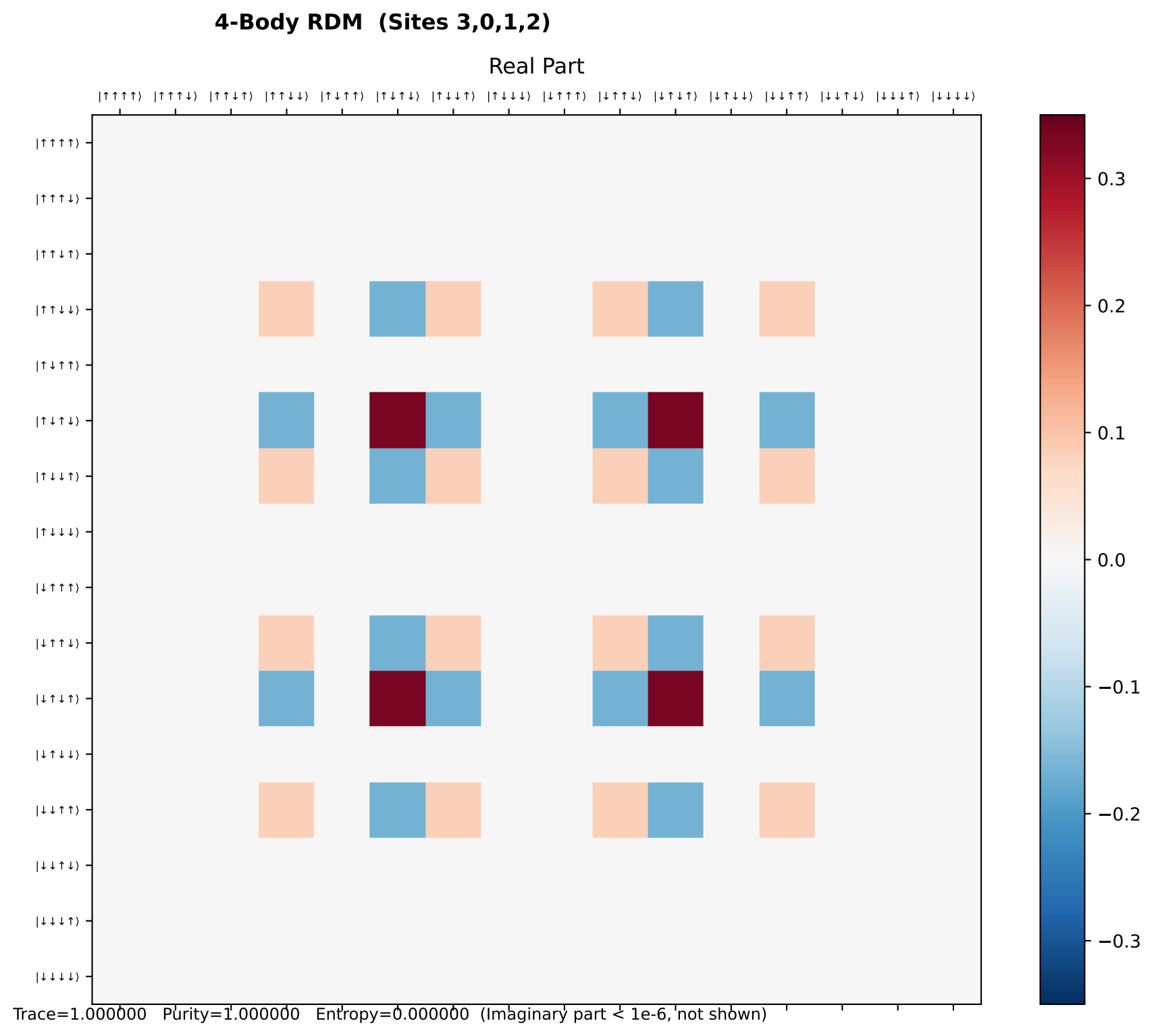
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 0, 1)	0.50000000
(2, 3)	(0, 1)	0.58333333
(2, 0)	(3, 1)	0.33333333
(2, 1)	(3, 0)	0.58333333
(2, 3, 0)	(1,)	0.50000000
(2, 3, 1)	(0,)	0.50000000
(2, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.


4-Body RDM (Sites 3,0,1,2)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 3,0,1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 1.0000000000
Von Neumann entropy	: 0.0000000000
Eigenvalue min	: -5.341574e-17
Eigenvalue max	: 1.000000e+00
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(0, 1, 2)	0.50000000
(3, 0)	(1, 2)	0.58333333
(3, 1)	(0, 2)	0.33333333
(3, 2)	(0, 1)	0.58333333
(3, 0, 1)	(2,)	0.50000000
(3, 0, 2)	(1,)	0.50000000
(3, 1, 2)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.